Deployment Guide

Recommended setup

This repository now includes a production-ready Flask backend in webbackend.py (/d:/Project/DyslexiaDetectionSystem/webbackend.py) and a Docker deployment definition in Dockerfile (/d:/Project/DyslexiaDetectionSystem/Dockerfile).

The deployment serves:

- the standalone browser dashboard from web/ (/d:/Project/DyslexiaDetectionSystem/web)
- the local transcription API at /api/reading-transcribe
- a health check at /healthz

Deployment Options

The project now includes configuration for three common hosting paths:

Platform	File	Best for
Render	[`render.yaml`](/d:/Project/Dyslexia_Detection_System/render.yaml)	Simple Docker deployment with automatic HTTPS
Railway	[`railway.json`](/d:/Project/Dyslexia_Detection_System/railway.json)	Fast app deployment from GitHub
Fly.io	[`fly.toml`](/d:/Project/Dyslexia_Detection_System/fly.toml)	Global app hosting with region control

Live Deployment

Current public demo:

- https://learning-disorder-intelligence.onrender.com/ (https://learning-disorder-intelligence.onrender.com/)

Deploy link:

- https://learning-disorder-intelligence.onrender.com/ (https://learning-disorder-intelligence.onrender.com/)

Render

Render is the simplest path if you want a quick public URL.

Steps:

1. Push this repository to GitHub.
1. Connect the repo to Render.
1. Let Render build the Docker image.

1. Open the public URL and verify /healthz returns ok.
1. Test the dashboard microphone workflow from the HTTPS URL.

Render auto-deploy troubleshooting

If the site does not update after a GitHub push, check these items in the Render dashboard:

1. The service is connected to archisman-das/Learning-Disorder-Intelligence.
1. The deploy branch is main.
1. Auto-deploy is enabled for the service.
1. The latest build completed successfully.
1. The health check path is still /healthz.

This repository is already configured for automatic deploys through render.yaml (/d:/Project/DyslexiaDetectionSystem/render.yaml), which sets autoDeploy: true and points Render at the Dockerized Flask backend.

Railway

Railway works well if you want a quick Git-based deploy with a Dockerfile.

Steps:

1. Push this repository to GitHub.
1. Create a Railway project from the repo.
1. Railway should use railway.json (/d:/Project/DyslexiaDetectionSystem/railway.json) and Dockerfile (/d:/Project/DyslexiaDetectionSystem/Dockerfile).
1. Confirm the service exposes the app on the assigned $PORT.
1. Visit /healthz and then open the public dashboard URL.

Fly.io

Fly is a strong choice if you want region-aware global deployment.

Steps:

1. Install flyctl.
1. Run fly launch or reuse the included fly.toml (/d:/Project/DyslexiaDetectionSystem/fly.toml).
1. Deploy with fly deploy.
1. Confirm the app is reachable over HTTPS and /healthz responds successfully.
1. Open the dashboard and test the microphone workflow.

Important notes

- Microphone access requires HTTPS or localhost.
- The transcription endpoint depends on openai-whisper and ffmpeg.
- The first audio request may be slower because the Whisper model is loaded on demand.
- The dashboard itself is mostly client-side, so browser state and records are stored locally in the user session unless you extend the backend further.
- The local web dashboard also includes a model statistics page that reads from stored comparison snapshots rather than live test labels.
- For Fly.io, the app is configured to listen on port 10000 and expose /healthz as a health check.

Local run

For local development:

```
python run_local_web.py
```

For production-like local testing:

```
gunicorn -b 0.0.0.0:10000 web_backend:app
```